

Education

- 2018 oct – 2023 jul 📖 **Ph.D. Engineering, University of Buenos Aires.**
Dissertation title: *Reconfigurable Networks of Unmanned Vehicles*.
- 2010 mar – 2017 jun 📖 **Electromechanical Engineering, National University of Mar del Plata.**
Integrated 6 year program, approximately equivalent to BSc + MSc.
Final project title: *Two-Wheeled Inverted Pendulum: A Robotic Prototype for Educational Purposes*.

Research Positions

Summary

My research lies at the intersection of **robotics and network control**, and is motivated by applications such as distributed robotic exploration, transportation, and human assistance. My work combines tools from **graph theory, control, optimization, and machine learning**, with the goal of developing **perception-and-interaction-aware coordination** strategies for heterogeneous multi-robot systems.

Postdoctoral Researcher

(Full-time independent postdoctoral research.)

- 2023 – **Present** 📖 **CONICET** (National Scientific and Technical Research Council, Argentina).
Host institution: University of Buenos Aires .
Instituto de Tecnologías y Ciencias de la Ingeniería “Hilario Fernández Long” (Institute of Engineering Technology and Sciences).
Research focus: *Graph rigidity theory and its applications to multi-robot control.*

Doctoral Researcher

(Full-time doctoral research leading to the award of the Ph.D.)

- 2018 – 2023 📖 **University of Buenos Aires** (Argentina).
Laboratorio de Automática y Robótica (Automation and Robotics Laboratory).
Research focus: *Scalable control methods for networks of unmanned vehicles.*

Research Publications

(*Corresponding author.)

Journal Articles

- 1 J. I. Alvarez-Hamelin, J. I. Giribet, I. Mas, and **J. F. Presenza***, “Extremal properties and bounds for the generalized algebraic connectivity of graphs in euclidean spaces,” *Linear Algebra and its Applications*, vol. 727, pp. 24–36, 2025, ISSN: 0024-3795. 🔗 DOI: <https://doi.org/10.1016/j.laa.2025.07.028>.
- 2 **J. F. Presenza***, I. Mas, J. Ignacio Alvarez-Hamelin, and J. I. Giribet, “Subframework-based bearing rigidity maintenance control in multirobot networks,” *IEEE Control Systems Letters*, vol. 9, pp. 1249–1254, 2025, Presented at the 2025 Conference on Decision and Control. 🔗 DOI: [10.1109/LCSYS.2025.3580768](https://doi.org/10.1109/LCSYS.2025.3580768).

- 3 P. Moreno, **J. F. Presenza**, I. Mas, and J. I. Giribet, "Aerial multi-camera robotic jib crane," *IEEE Robotics and Automation Letters*, vol. 6, no. 2, pp. 4103–4108, 2021.  DOI: 10.1109/LRA.2021.3065299.
- 4 C. Pose, **F. Presenza**, I. Mas, and J. I. Giribet, "Trajectory following with a mav under rotor fault conditions," *Unmanned Systems*, vol. 08, no. 04, pp. 263–268, 2020.  DOI: 10.1142/S2301385020500193.

Conference Proceedings

- 1 E. Pecker Marcosig, **J. F. Presenza**, I. Mas, J. I. Alvarez-Hamelin, J. I. Giribet, and R. Castro, "Simulation of decentralized coordination strategies for networked multi-robot systems with emergent behavior-devs," in *Proceedings of the 2025 Winter Simulation Conference*, 2025.  DOI: 10.1109/WSC68292.2025.11338929.
- 2 **J. F. Presenza***, J. I. Alvarez-Hamelin, I. Mas, and J. I. Giribet, "Subframework-based rigidity control in multirobot networks," in *2022 American Control Conference (ACC)*, 2022, pp. 3431–3436.  DOI: 10.23919/ACC53348.2022.9867693.

Books and Chapters

- 1 **J. F. Presenza***, E. Pecker Marcosig, I. Mas, J. I. Alvarez-Hamelin, and J. I. Giribet, *Scalable Rigidity Maintenance Strategies for Networked Multirobot Systems: Overcoming Decentralized Coordination Challenges*. 2025, Book chapter accepted for publication on Modeling, Simulation and Visualisation-Driven study of Resilient Complex Adaptive Systems, Wiley.

Refereed Conference Papers

- 1 **J. F. Presenza**, I. Mas, J. I. Giribet, and J. I. Alvarez-Hamelin, "Descomposición óptima en subframeworks con aplicaciones al control de redes multirobot," XII Jornadas Argentinas de Robótica, 2024.
- 2 **J. F. Presenza**, J. I. Giribet, J. I. Alvarez-Hamelin, and I. Mas, "Control de rigidez por subframeworks en sistemas multirobot," XI Jornadas Argentinas de Robótica, 2022.
- 3 E. Pecker Marcosig, **J. F. Presenza**, J. I. Giribet, and I. Mas, "Sistema de control para un robot aerodeslizante," X Jornadas Argentinas de Robótica, 2019.
- 4 **J. F. Presenza**, P. Claudio, I. Mas, and J. I. Giribet, "Validación de un sistema de posicionamiento ultrasónico para interiores," X Jornadas Argentinas de Robótica, 2019.
- 5 **J. F. Presenza**, E. M. Gelos, S. A. González, and J. R. Fischer, "Plataforma móvil basada en un péndulo invertido de dos ruedas," XVII Reunión de Trabajo sobre Procesamiento de la Información y Control, 2017.

Submitted Manuscripts

- 1 **J. F. Presenza***, L. J. Colombo, J. I. Giribet, and I. Mas, *Angle-based localization and rigidity maintenance control for multi-robot networks*, under revision. Preprint available at <https://arxiv.org/abs/2604.11754>, 2026.

Teaching Positions

Summary

Over **10 years of university-level teaching** in mathematics and engineering across public and private universities. Extensive experience in undergraduate instruction, course coordination and assessment design, with an estimated 2000 contact hours of classroom teaching.

Lecturer (part-time) — Course Instructor of Record

(Full responsibility for course design, lectures, assessment, and grading. ~ 60 contact hours per course edition.)

- 2025  **Harmonic Analysis** (Undergraduate - Engineering).
Department of Engineering, University of San Andrés, Argentina.
1 semester-long course.
- 2024 – 2025  **Discrete Mathematics** (Undergraduate - Engineering).
Department of Engineering, University of San Andrés, Argentina.
3 semester-long courses.

Lecturer (part-time) — Tutorial Section Instructor

(Regular tutorial lecturing and collaboration on course design, assessment, and grading. ~ 60 contact hours per course edition.)

- 2019 – 2025  **Linear Algebra** (Undergraduate - Engineering).
Department of Mathematics, University of Buenos Aires, Argentina.
13 semester-long courses.
- 2021 – 2024  **Discrete Mathematics** (Undergraduate - Engineering).
Department of Engineering, University of San Andrés, Argentina.
6 semester-long courses.
- 2021  **Linear Algebra** (Undergraduate - Engineering).
Department of Engineering, University of San Andrés, Argentina.
1 semester-long course.

Teaching Assistant (Part-time)

(Regular tutorial lecturing and collaboration on course content. ~ 60 contact hours per course edition.)

- 2013 – 2016  **Engineering Mathematics** (Undergraduate - Engineering).
Department of Mathematics, National University of Mar del Plata, Argentina.
8 semester-long courses.

Funded Research Projects and Fellowships

Research Projects

- 2023 - Present  **University of Buenos Aires** (Internal research grant).
Project title: *Dynamics of Information Flow on the Internet and Social Networks from a Complex Systems Perspective*.
Role: Postdoctoral researcher. P.I.: J. Ignacio Alvarez-Hamelin.

Fellowships

- 2023 - Present  **Postdoctoral Fellowship**, CONICET.
Individually awarded competitive national fellowship.

Funded Research Projects and Fellowships (continued)

2018 - 2023

- **Doctoral Fellowship**, University of Buenos Aires.
Individually awarded competitive national fellowship.

Supervision

Undergraduate thesis

- Ongoing ■ **Manuel Nuñez, co-supervised with Dr. Ezequiel Pecker-Marcosig**
Engineering degree: 6 year program, approximately equivalent to BSc + MSc.
Thesis title: *Hybrid DEVS and ROS2 Embedded Simulation for Multi-Robot Formations*
- 2024 ■ **Eng. Luana Quirolo, co-supervised with Dr. Juan Giribet.**
Engineering degree: 6 year program, approximately equivalent to BSc + MSc.
Thesis title: *Robot Velocity Estimation with Optical Flow Sensors*

Research and Educational Software

- **Rigidity-based Multi-Robot Coordination.**
https://github.com/fpresenza/uv_network
- **Event-based Simulation Library for Dynamical Multi-Agent Systems.**
<https://github.com/fpresenza/EBDEVS-multiagent>
- **Applied Mathematics for Educational Purposes.**
<https://github.com/fpresenza/appliedmath>
- **Control and Estimation of an Omnidirectional Mobile Robot with Arduino and ROS.**
<https://github.com/fpresenza/omniduino>
- **Rust-based Cryptographic Tools.**
https://github.com/fpresenza/sparkling_water_bootcamp

Skills

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|----------------------|---|
| Languages | ■ Spanish: Native proficiency. |
| | ■ English: Proficient (C1/C2 level, CEFR) (5 year coursework and professional use). |
| | ■ French: Elementary (A2 level, CEFR) (1 year coursework). |
| Programming | ■ Python, C/C++, Rust, R, MATLAB, Git, $\LaTeX$. |
| Scientific Computing | ■ SageMath, NetworkX, Graph-tool, PowerDEVS, PyPDEVS |
| Robotics Tools | ■ ROS/ROS2, Gazebo, PX4, Raspberry Pi, Arduino. |

Awards and Honors

- 2023 ■ **Ph.D. Thesis evaluated as “sobresaliente” (outstanding)** by the examination committee, University of Buenos Aires.
- 2017 ■ **Graduated with Diploma of Honor (average grade: 8.67/10)**, National University of Mar del Plata.

Certification

Languages

- 2013  **French Language – 1st Year Completed (approx. CEFR level A2).** Laboratorio de Idiomas, National University of Mar del Plata.
- 2012  **English Language – 5th Year Completed (approx. CEFR level C1/C2).** Laboratorio de Idiomas, National University of Mar del Plata.

Advanced Coursework

- 2024  **Applied Cryptography**, Lambda Class (<https://lambdaclass.com/>).
- 2021  **Convex Optimization**, University of Buenos Aires.
- 2020  **Statistical Learning**, University of Buenos Aires.
- 2019  **Robust Control Theory**, University of Buenos Aires.
-  **Integrated Navigation Systems**, University of Buenos Aires.
-  **Mobile Robotics: A Probabilistic Approach**, University of Buenos Aires.

Additional Professional Experience

- 2017 jul – 2018 sep  **Junior Professional Program**, Trenes Argentinos (National railway operator).